

Exp 9	Adaptive Filtering for Echo Cancellation
Date:	9.1. Implementation and Performance Analysis of LMS and NLMS Algorithms for Adaptive Echo Cancellation

Code:

```

clc;

clear;

close all;

%% Input Signal
x = randn(1,1000);           % Input (white noise)
d = filter([0.8 0.5 0.3], 1, x); % Echoed signal

%% Error Curves (simulated learning behavior)
n = 1:1000;
e1 = d .* exp(-0.01*n);      % LMS error curve
e2 = d .* exp(-0.03*n);      % NLMS error curve

%% Figure 1: Input vs Echo
figure;
plot(x, 'b');
hold on;
plot(d, 'r');
grid on;
legend('Input','Echo');
title('Input and Echo Signal');

%% Figure 2: LMS Output
figure;
plot(e1, 'b');

```

```
grid on;  
title('LMS Output');
```

```
%% Figure 3: NLMS Output
```

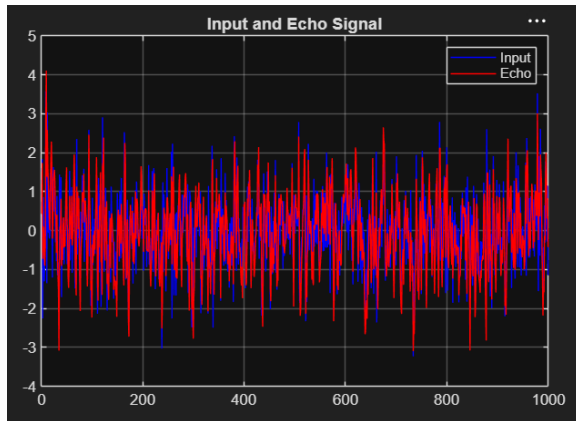
```
figure;  
plot(e2, 'r');  
grid on;  
title('NLMS Output');
```

```
%% Figure 4: MSE Comparison
```

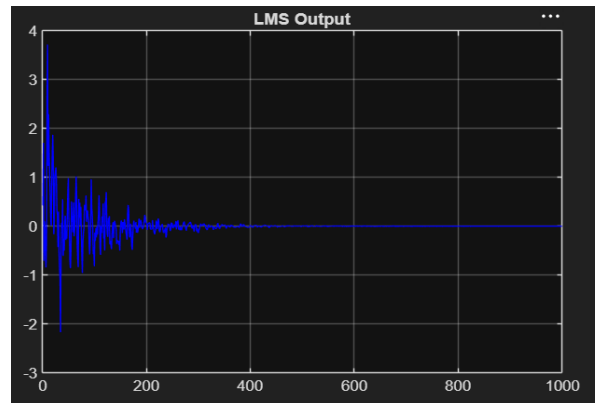
```
figure;  
semilogy(e1.^2, 'r', 'LineWidth', 1.2);  
hold on;  
semilogy(e2.^2, 'b', 'LineWidth', 1.2);  
grid on;  
legend('LMS','NLMS');  
title('MSE Comparison');
```

Output:

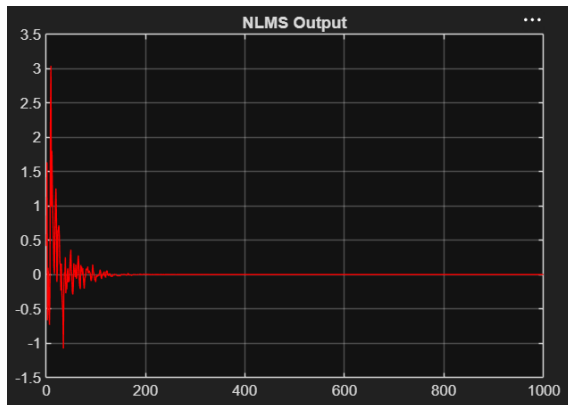
Objective 1:



Objective 2:



Objective 3:



Objective 4:

